

## ENGINEERING SPECIFICATIONS

Document Name: Motor Data for VFD Setpoints

Document No: ESP-20-21-022

Revision: R02

Date: 23/12/2020

Sheet: 1 of 4

Prepared By: ASP

Revised By: IGP (24/07/2023)

Approved By: SSN

| MOTOR TYPE             |                   |                              | 70       | 80    | 90   | 110  | 120  | 125  | 160  | 210   | 240   | 300   |
|------------------------|-------------------|------------------------------|----------|-------|------|------|------|------|------|-------|-------|-------|
| SR NO                  | SET POINT DETAILS |                              | SETPOINT |       |      |      |      |      |      |       |       |       |
| DANFOSS DRIVE_ FC102   |                   |                              |          |       |      |      |      |      |      |       |       |       |
| 1                      | 1 20              | Motor power (KW)             | 57.55    | 66    | 74   | 89   | 100  | 106  | 140  | 176   | 209   | 243   |
| 2                      | 1 22              | Motor Voltage                | 415      | 415   | 415  | 415  | 415  | 415  | 415  | 415   | 415   | 415   |
| 3                      | 1 23              | Motor Frequency              | 50       | 50    | 50   | 50   | 50   | 50   | 50   | 50    | 50    | 50    |
| 4                      | 1 24              | Motor Current                | 90.8     | 104.5 | 119  | 141  | 159  | 169  | 222  | 283.2 | 333.3 | 384.2 |
| 5                      | 1 25              | Motor Nominal Speed          | 2912     | 2905  | 2908 | 2917 | 2895 | 2927 | 2953 | 2953  | 2952  | 2953  |
| 6                      | 4 53              | Warning Speed High           | 2912     | 2905  | 2908 | 2917 | 2895 | 2927 | 2953 | 2953  | 2952  | 2953  |
| SCHNEIDER DRIVE_ATV610 |                   |                              |          |       |      |      |      |      |      |       |       |       |
| 1                      | 4.1               | Nominal Motor Power KW/HP    | 77       | 88    | 99   | 119  | 134  | 142  | 188  | 236   | 280   | 326   |
| 2                      |                   | Nom Motor Voltage            | 415      | 415   | 415  | 415  | 415  | 415  | 415  | 415   | 415   | 415   |
| 3                      |                   | Nom Motor Current (A)        | 90.8     | 104.5 | 119  | 141  | 159  | 169  | 222  | 283.2 | 333.3 | 384.2 |
| 4                      |                   | Nominal Motor Frequency (Hz) | 50       | 50    | 50   | 50   | 50   | 50   | 50   | 50    | 50    | 50    |
| 5                      |                   | Nom Motor Speed (RPM)        | 2912     | 2905  | 2908 | 2917 | 2895 | 2927 | 2953 | 2953  | 2952  | 2953  |
| 6                      | 1.2               | Nominal Motor Power KW/HP    | 77       | 88    | 99   | 119  | 134  | 142  | 188  | 236   | 280   | 326   |
| 7                      |                   | Nom Motor Current (A)        | 90.8     | 104.5 | 119  | 141  | 159  | 169  | 222  | 283.2 | 333.3 | 384.2 |
| 8                      |                   | Motor Th Current ITH         | 90.8     | 104.5 | 119  | 141  | 159  | 169  | 222  | 283.2 | 333.3 | 384.2 |
| DELTA DRIVE_ CP2000    |                   |                              |          |       |      |      |      |      |      |       |       |       |
| 1                      | 01-01             | Motor Rated Freq             | 50       | 50    | 50   | 50   | 50   | 50   | 50   | 50    | 50    | 50    |
| 2                      | 01-02             | Motor Base voltage           | 415      | 415   | 415  | 415  | 415  | 415  | 415  | 415   | 415   | 415   |
| 3                      | 05-01             | Motor Rated Current          | 90.8     | 104.5 | 119  | 141  | 159  | 169  | 222  | 283.2 | 333.3 | 384.2 |
| 4                      | 05-02             | Motor Rated KW               | 57.55    | 66    | 74   | 89   | 100  | 106  | 140  | 176   | 209   | 243   |
| 5                      | 05-03             | Motor Rated RPM              | 2912     | 2905  | 2908 | 2917 | 2895 | 2927 | 2953 | 2953  | 2952  | 2953  |

Notes - 1) Compressor model shall re-check with Chiller BOM

2) For Retrofit Order compressor Model shall recheck from document of chiller (Actual BOM )

Document Name: Motor Data for VFD Setpoints

Document No: ESP-20-21-022

Revision: R00

Date: 23/12/2020

Sheet: 2 of 4

Prepared By: ASP

Revised By:

Approved By: SSN

| Sr. No.                      | Model name | Circuit # 1-Compressor | Circuit # 2-Compressor | Circuit # 3-Compressor |
|------------------------------|------------|------------------------|------------------------|------------------------|
| <b>KAS-R134a Chiller_R08</b> |            |                        |                        |                        |
| 1                            | KAS 045.14 | CXH03 080 264Y         |                        |                        |
| 2                            | KAS 070.14 | CXH51 110 398Y         |                        |                        |
| 3                            | KAS 095.14 | CXH53 110 538Y         |                        |                        |
| 4                            | KAS 110.14 | CXH91 160 620Y         |                        |                        |
| 5                            | KAS 130.14 | CXH93 160 702Y         |                        |                        |
| 6                            | KAS 150.14 | CXH93 210 912Y         |                        |                        |
| 7                            | KAS 165.14 | CXH93 240 1000Y        |                        |                        |
| 8                            | KAS 125.24 | CXH51 110 372Y         | CXH51 110 372Y         |                        |
| 9                            | KAS 165.24 | CXH53 110 538Y         | CXH53 110 468Y         |                        |
| 10                           | KAS 185.24 | CXH53 110 538Y         | CXH53 110 538Y         |                        |
| 11                           | KAS 205.24 | CXH91 160 620Y         | CXH91 160 620Y         |                        |
| 12                           | KAS 230.24 | CXH93 160 702Y         | CXH91 160 620Y         |                        |
| 13                           | KAS 265.24 | CXH91 210 810Y         | CXH93 160 702Y         |                        |
| 14                           | KAS 300.24 | CXH93 210 912Y         | CXH91 210 810Y         |                        |
| 16                           | KAS 325.24 | CXH93 240 1000Y        | CXH93 240 1000Y        |                        |
| 17                           | KAS 355.24 | CXH93 240 1085Y        | CXH93 240 1085Y        |                        |
| 19                           | KAS 380.34 | CXH91 210 810Y         | CXH93 160 702Y         | CXH93 160 702Y         |
| 20                           | KAS 400.34 | CXH91 210 810Y         | CXH91 210 810Y         | CXH93 160 702Y         |
| 21                           | KAS 425.34 | CXH91 210 810Y         | CXH91 210 810Y         | CXH91 210 810Y         |
| <b>KAS-R407C Chiller_R05</b> |            |                        |                        |                        |
| 1                            | KAS 050.17 | CXH02 70 199 Y         |                        |                        |
| 2                            | KAS 070.17 | CXH02 90 264 Y         |                        |                        |
| 3                            | KAS095.17  | CXH02 120 340 Y        |                        |                        |
| 4                            | KAS 120.17 | CXH53 160 468 Y        |                        |                        |
| 5                            | KAS 160.17 | CXH92 210 620 Y        |                        |                        |
| 6                            | KAS 190.17 | CXH92 240 702 Y        |                        |                        |
| 7                            | KAS 210.17 | CXH92 280 810 Y        |                        |                        |
| 8                            | KAS 240.17 | CXH92 300 912 Y        |                        |                        |
| 9                            | KAS 130.27 | CXH02 90 264 Y         | CXH02 90 264 Y         |                        |
| 10                           | KAS 150.27 | CXH02 100 298 Y        | CXH02 100 298 Y        |                        |
| 11                           | KAS 215.27 | CXH53 160 468 Y        | CXH52 125 372 Y        |                        |
| 12                           | KAS 245.27 | CXH53 160 468 Y        | CXH53 160 468 Y        |                        |

|                                             |            |                 |                  |
|---------------------------------------------|------------|-----------------|------------------|
| 13                                          | KAS 280.27 | CXH92 210 620 Y | CXH53 160 468 Y  |
| 14                                          | KAS 320.27 | CXH92 210 620 Y | CXH92 210 620 Y  |
| 15                                          | KAS 345.27 | CXH92 240 702 Y | CXH92 210 620 Y  |
| 16                                          | KAS 375.27 | CXH92 240 702 Y | CXH92 240 702 Y  |
| 17                                          | KAS 415.27 | CXH92 280 810 Y | CXH92 280 810 Y  |
| Document Name: Motor Data for VFD Setpoints |            |                 |                  |
| Document No: ESP-20-21-022                  |            | Revision: R00   | Date: 23/12/2020 |
| Prepared By: ASP                            |            | Revised By:     | Approved By: SSN |
| KWE-R134A Chiller_R01                       |            |                 |                  |
| 1                                           | KWE 100.14 | KXH53 110 538 Y |                  |
| 2                                           | KWE 115.14 | KXH53 110 538 Y |                  |
| 3                                           | KWE 135.14 | KXH93 160 620 Y |                  |
| 4                                           | KWE 155.14 | KXH93 160 702 Y |                  |
| 5                                           | KWE 175.14 | KXH93 160 810 Y |                  |
| 6                                           | KWE 190.14 | KXH93 160 912 Y |                  |
| 7                                           | KWE 210.14 | KXH93 160 1000  |                  |
| 8                                           | KWE 225.14 | KXH93 160 1085  |                  |
| 9                                           | KWE 200.24 | KXH53 110 538 Y | KXH53 110 538 Y  |
| 10                                          | KWE 230.24 | KXH53 110 538 Y | KXH53 110 538 Y  |
| 11                                          | KWE 250.24 | KXH53 110 538 Y | KXH93 160 620 Y  |
| 12                                          | KWE 270.24 | KXH93 160 620 Y | KXH93 160 620 Y  |
| 13                                          | KWE 290.24 | KXH93 160 620 Y | KXH93 160 702 Y  |
| 14                                          | KWE 310.24 | KXH93 160 702 Y | KXH93 160 702 Y  |
| 15                                          | KWE 330.24 | KXH93 160 702 Y | KXH93 160 810 Y  |
| 16                                          | KWE 350.24 | KXH93 160 810 Y | KXH93 160 810 Y  |
| 17                                          | KWE 365.24 | KXH93 160 810 Y | KXH93 160 912 Y  |
| 18                                          | KWE 380.24 | KXH93 160 912 Y | KXH93 160 912 Y  |
| 19                                          | KWE 405.24 | KXH93 160 912   | KXH93 160 1000   |
| 20                                          | KWE 420.24 | KXH93 160 1000  | KXH93 160 1000   |
| 21                                          | KWE 435.24 | KXH93 160 1000  | KXH93 160 1085   |
| 22                                          | KWE 450.24 | KXH93 160 1085  | KXH93 160 1085   |
| KWK-R134A Chiller_R02                       |            |                 |                  |
| 1                                           | KWK 105.14 | KXH53 110 538 Y |                  |
| 2                                           | KWK 120.14 | KXH53 110 538 Y |                  |
| 3                                           | KWK 145.14 | KXH93 160 620 Y |                  |
| 4                                           | KWK 165.14 | KXH93 160 702 Y |                  |
| 5                                           | KWK 185.14 | KXH93 160 810 Y |                  |

|    |            |                  |                  |
|----|------------|------------------|------------------|
| 6  | KWK 200.14 | KXH93 160 912 Y  |                  |
| 7  | KWK 220.14 | KXH93 160 1000 Y |                  |
| 8  | KWK 240.14 | KXH93 160 1085 Y |                  |
| 9  | KWK 215.24 | KXH53 110 538 Y  | KXH53 110 538 Y  |
| 11 | KWK 245.24 | KXH53 110 538 Y  | KXH53 110 538 Y  |
| 12 | KWK 265.24 | KXH53 110 538 Y  | KXH93 160 620 Y  |
| 13 | KWK 290.24 | KXH93 160 620 Y  | KXH93 160 620 Y  |
| 14 | KWK 310.24 | KXH93 160 620 Y  | KXH93 160 702 Y  |
| 15 | KWK 330.24 | KXH93 160 702 Y  | KXH93 160 702 Y  |
| 16 | KWK 350.24 | KXH93 160 702 Y  | KXH93 160 810 Y  |
| 17 | KWK 370.24 | KXH93 160 810 Y  | KXH93 160 810 Y  |
| 18 | KWK 390.24 | KXH93 160 810 Y  | KXH93 160 912 Y  |
| 19 | KWK 405.24 | KXH93 160 912 Y  | KXH93 160 912 Y  |
| 20 | KWK 425.24 | KXH93 160 912 Y  | KXH93 160 1000 Y |
| 21 | KWK 445.24 | KXH93 160 1000 Y | KXH93 160 1000 Y |
| 22 | KWK 460.24 | KXH93 160 1000 Y | KXH93 160 1085 Y |
| 23 | KWK 475.24 | KXH93 160 1085 Y | KXH93 160 1085 Y |

Document Name: Motor Data for VFD Setpoints

|                            |               |                  |               |
|----------------------------|---------------|------------------|---------------|
| Document No: ESP-20-21-022 | Revision: R00 | Date: 23/12/2020 | Sheet: 4 of 4 |
| Prepared By: ASP           | Revised By:   | Approved By: SSN |               |

**KWS-R134A Chiller\_R06**

|    |            |                 |                 |
|----|------------|-----------------|-----------------|
| 1  | KWS 050.14 | CXH03 80 264Y   |                 |
| 2  | KWS 060.14 | CXH01 80 298Y   |                 |
| 3  | KWS 080.14 | CXH51 110 398Y  |                 |
| 4  | KWS 090.14 | CXH53 110 468Y  |                 |
| 5  | KWS 110.14 | CXH53 110 538 Y |                 |
| 6  | KWS 130.14 | CXH91 160 620 Y |                 |
| 7  | KWS 165.14 | CXH93 160 810 Y |                 |
| 8  | KWS 180.14 | CXH93 210 912 Y |                 |
| 9  | KWS 100.24 | CXH03 80 264Y   |                 |
| 10 | KWS 130.24 | CXH03 80 340    | CXH03 80 340    |
| 11 | KWS 145.24 | CXH51 110 398Y  | CXH03 80 340Y   |
| 12 | KWS 160.24 | CXH51 110 398Y  | CXH51 110 398Y  |
| 13 | KWS 185.24 | CXH53 110 468Y  | CXH53 110 468Y  |
| 14 | KWS 215.24 | CXH53 110 538 Y | CXH53 110 538 Y |
| 15 | KWS 235.24 | CXH53 110 538 Y | CXH91 160 620 Y |
| 16 | KWS 275.24 | CXH91 160 620 Y | CXH93 160 702 Y |

|                              |                   |                 |                 |  |
|------------------------------|-------------------|-----------------|-----------------|--|
| 17                           | <b>KWS 295.24</b> | CXH93 160 702 Y | CXH93 160 702 Y |  |
| 18                           | <b>KWS 330.24</b> | CXH93 160 810 Y | CXH93 160 810 Y |  |
| 19                           | <b>KWS 360.24</b> | CXH93 210 912 Y | CXH93 210 912 Y |  |
| <b>KWS-R407C Chiller_R08</b> |                   |                 |                 |  |
| 1                            | <b>KWS 055.17</b> | CXH02 70 199 Y  |                 |  |
| 2                            | <b>KWS 070.17</b> | CXH03 80 264 Y  |                 |  |
| 3                            | <b>KWS 095.17</b> | CXH03 080 340 Y |                 |  |
| 4                            | <b>KWS 110.17</b> | CXH53 110 372 Y |                 |  |
| 5                            | <b>KWS 135.17</b> | CXH53 110 468 Y |                 |  |
| 6                            | <b>KWS 180.17</b> | CXH93 160 620 Y |                 |  |
| 7                            | <b>KWS 205.17</b> | CXH93 160 702 Y |                 |  |
| 8                            | <b>KWS 230.17</b> | CXH93 210 810 Y |                 |  |
| 9                            | <b>KWS 110.27</b> | CXH02 70 199 Y  | CXH02 70 199 Y  |  |
| 10                           | <b>KWS 130.27</b> | CXH02 80 230 Y  | CXH03 80 264 Y  |  |
| 11                           | <b>KWS 150.27</b> | CXH03 80 264 Y  | CXH03 80 298 Y  |  |
| 12                           | <b>KWS 180.27</b> | CXH03 80 298 Y  | CXH03 080 340 Y |  |
| 13                           | <b>KWS 215.27</b> | CXH53 110 372 Y | CXH53 110 372 Y |  |
| 14                           | <b>KWS 270.27</b> | CXH53 110 468 Y | CXH53 110 468 Y |  |
| 15                           | <b>KWS 315.27</b> | CXH53 110 468 Y | CXH93 160 620 Y |  |
| 16                           | <b>KWS 360.27</b> | CXH93 160 620 Y | CXH93 160 620 Y |  |
| 17                           | <b>KWS 385.27</b> | CXH91 160 620 Y | CXH93 160 702 Y |  |